

APSPDCL · COMMERCIAL ROOFTOP

200 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

Net investment ₹81.90 L after input tax credit, generating 3,06,600 units in year one against an energy charge of ₹7.65 per unit.

PAYBACK 3.9 yrs Post-tax, undiscounted	PROJECT IRR 26.2% Unlevered, 30 years	COST OF SOLAR ENERGY ₹3.32 per kWh, against ₹7.65 from the grid	NET PRESENT VALUE ₹1.39 Cr Discounted at 10%
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PLANT AND GENERATION	INVESTMENT AND TAX BASIS																																
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ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	81,90,476	32,76,190	49,14,286	8,24,551
2	49,14,285	19,65,714	29,48,571	4,94,731
3	29,48,572	11,79,429	17,69,143	2,96,839
4	17,69,143	7,07,657	10,61,486	1,78,103
5	10,61,485	4,24,594	6,36,891	1,06,862
6	6,36,892	2,54,757	3,82,135	64,117
7	3,82,135	1,52,854	2,29,281	38,470
8	2,29,281	91,712	1,37,569	23,082
9	1,37,568	55,027	82,541	13,849
10	82,541	33,016	49,525	8,309
11	49,525	19,810	29,715	4,986
12	29,715	11,886	17,829	2,991
Tax shield, years 1-12				20,56,892

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

200 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	3,06,600	21,95,256	—	32,76,190	(2,72,050)	24,67,306	24,67,306
2	3,05,067	22,71,651	1,20,000	19,65,714	46,797	21,04,854	45,72,160
3	3,03,542	23,50,704	1,20,000	11,79,429	2,64,585	19,66,119	65,38,279
4	3,02,024	24,32,509	1,20,000	7,07,657	4,03,909	19,08,600	84,46,879
5	3,00,514	25,17,160	1,20,000	4,24,594	4,96,455	19,00,705	1,03,47,584
6	2,99,011	26,04,757	1,20,000	2,54,757	5,61,247	19,23,511	1,22,71,095
7	2,97,516	26,95,403	1,20,000	1,52,854	6,09,707	19,65,696	1,42,36,791
8	2,96,029	27,89,203	1,20,000	91,712	6,48,703	20,20,500	1,62,57,291
9	2,94,548	28,86,267	1,20,000	55,027	6,82,365	20,83,902	1,83,41,193
10	2,93,076	29,86,709	1,20,000	33,016	7,13,184	21,53,525	2,04,94,718
11	2,91,610	30,90,647	1,20,000	19,810	7,42,667	22,27,980	2,27,22,698
12	2,90,152	31,98,201	1,20,000	11,886	7,71,730	23,06,471	2,50,29,169
13	2,88,702	33,09,499	1,20,000	7,132	8,00,938	23,88,561	2,74,17,730
14	2,87,258	34,24,669	1,20,000	4,279	8,30,642	24,74,027	2,98,91,757
15	2,85,822	35,43,848	1,20,000	2,567	8,61,068	25,62,780	3,24,54,537
16	2,84,393	36,67,174	1,20,000	1,540	8,92,365	26,54,809	3,51,09,346
17	2,82,971	37,94,791	1,20,000	924	9,24,639	27,50,152	3,78,59,498
18	2,81,556	39,26,850	1,20,000	555	9,57,968	28,48,882	4,07,08,380
19	2,80,148	40,63,504	1,20,000	333	9,92,417	29,51,087	4,36,59,467
20	2,78,747	42,04,914	1,20,000	200	10,28,041	30,56,873	4,67,16,340
21	2,77,354	43,51,245	1,20,000	120	10,64,890	31,66,356	4,98,82,696
22	2,75,967	45,02,669	1,20,000	72	11,03,012	32,79,657	5,31,62,353
23	2,74,587	46,59,362	1,20,000	43	11,42,456	33,96,906	5,65,59,259
24	2,73,214	48,21,507	1,20,000	26	11,83,269	35,18,238	6,00,77,497
25	2,71,848	49,89,296	1,20,000	16	12,25,500	36,43,795	6,37,21,292
26	2,70,489	51,62,923	1,20,000	9	12,69,201	37,73,723	6,74,95,015
27	2,69,136	53,42,593	1,20,000	6	13,14,421	39,08,172	7,14,03,187
28	2,67,791	55,28,515	1,20,000	3	13,61,214	40,47,301	7,54,50,488
29	2,66,452	57,20,908	1,20,000	2	14,09,636	41,91,272	7,96,41,760
30	2,65,119	59,19,995	1,20,000	1	14,59,742	43,40,253	8,39,82,013
Total	85,61,243	11,29,52,729	34,80,000	81,90,474	2,54,90,718	8,39,82,013	8,39,82,013

RETURN METRICS

Simple payback	3.9 yrs
Discounted payback	5.2 yrs
Project IRR, unlevered	26.2%
Net present value at 10%	₹1.39 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.32 / kWh
Cumulative return, 30 years	925%
Total net cash flow, after tax	₹8.40 Cr
30-year generation	85,61,241 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.32
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.32 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 85,61,241 units, avoiding 7,020.2 tonnes of CO₂.

ASSUMPTIONS

- Tariff:

HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.

Demand charges:

Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.

Tariff escalation:

4% per year, applied to both the energy charge and the export rate.

Generation:

1,533 kWh per kWp in year one, degrading 0.5% per year.

Unit split:

90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.

GST:

5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹86.00 L.

Depreciation:

40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.

Tax:

25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:

Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.

Commissioning:

Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.

Maintenance:

₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.

Inverter replacement:

Not modelled. A mid-life inverter replacement would reduce the figures shown.

Financing:

None. All returns are unlevered — the return on the money invested, before any loan.

Horizon:

30 years, which runs beyond the module performance warranty.

Discount rate:

10% for net present value, discounted payback and levelised cost.

Emissions:

Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 200 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.